

a**Fixed analysis inputs**

Same dataset

Same preprocessing

Same requested K

Same method settings

Different random seed

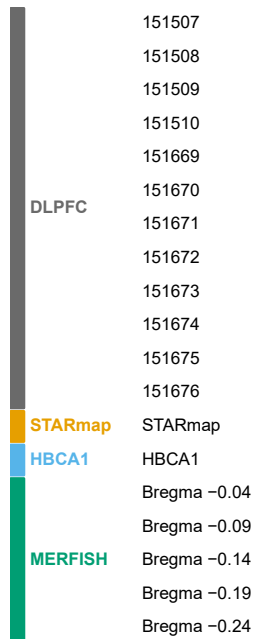
**Evaluated separately**

Reference accuracy

Partition reproducibility

Iso-accuracy divergence

Marker reproducibility

b**Benchmark landscape****c****Complete 20-seed coverage**

151507	20	20	20	20	20
151508	20	20	20	20	20
151509	20	20	20	20	20
151510	20	20	20	20	20
151669	20	20	20	20	20
151670	20	20	20	20	20
151671	20	20	20	20	20
151672	20	20	20	20	20
151673	20	20	20	20	20
151674	20	20	20	20	20
151675	20	20	20	20	20
151676	20	20	20	20	20
STARmap	20	20	20	20	20
HBCA1	20	20	20	20	20
Bregma -0.04	20	20	20	20	20
Bregma -0.09	20	20	20	20	20
Bregma -0.14	20	20	20	20	20
Bregma -0.19	20	20	20	20	20
Bregma -0.24	20	20	20	20	20

GraphST
STAGATE
SpaGCN
BANKSY
SEDR

20 seeds per method-dataset unit
95 units; 1,900 runs